However, ForecastBench is a relatively easy forecasting benchmark. On the more challenging MarketLiquid benchmark, the AIA Forecaster trails the performance of the market consensus. However, as we discuss in Section 5.3, the AIA Forecaster is *diversifying* to this consensus, and an ensemble of the two outperforms either the market consensus or the AIA Forecaster considered separately. We now turn to arguably the most important component of a high-quality forecasting system: the search pipeline.

## 5 Understanding Search in Forecasting

Search is a critical component of human forecasting, as timely news affects the likelihood of an event occurring. Efficient markets, by definition, aggregate salient information into market prices (Fama, 1970). We can see this effect in public prediction markets, which offer contracts that pay a fixed amount if an event occurs (and, typically, a separate contract which pays a fixed amount if it doesn't occur). This allows us to directly observe the market-implied odds of an event occurring.[8] For example, consider the question: will Grok 4 be released before the end of 2025? On June 27, 2025, Elon Musk announced that Grok 4 would be released just after July 4, 2025. This news announcement caused the market-implied odds to jump to 95-99%, with the market resolving to yes on July 22. We show the market price movement in Figure 2.

However, prior work for LLM forecasters suggests that search does not affect LLM forecasting performance! For example, Karger et al. (2024) find that the best-performing LLM-based forecaster omits search entirely, and adding a search pipeline actually *reduces* forecasting accuracy. Schoenegger et al. (2024) similarly do not find evidence that access to internet search improves forecasting performance.

In this section, we show that advanced forms of search are indeed critical for performance on a wide range of markets and that the structure of prior LLM forecasting benchmarks strongly affects the ability of search to help LLM forecasters.

To do so, we must first deal with the challenge of *foreknowledge bias*. All of the benchmarks described in Section 4 consist entirely of events which have already occurred, which enables us to evaluate performance by comparing forecasts to ground truth outcomes. Of course, this evaluation is rendered moot if we are unable to simulate how the forecaster would have behaved in the past, by blinding it to e.g., contemporaneous news articles which describe the event in question.[9] We show that this is a surprisingly challenging problem and detail our mitigation strategies.

8. We use "market price" and "market-implied odds" interchangeably, ignoring transaction costs, margin requirements, and other aspects of market structure. A trader seeking to profit from informed forecasts would of course not have this luxury.
9. Sarkar and Vafa (2024) describe foreknowledge acquired via pre-training as "lookahead bias." We use the term foreknowledge bias to include other forms of foreknowledge, including information that is exposed via search results.

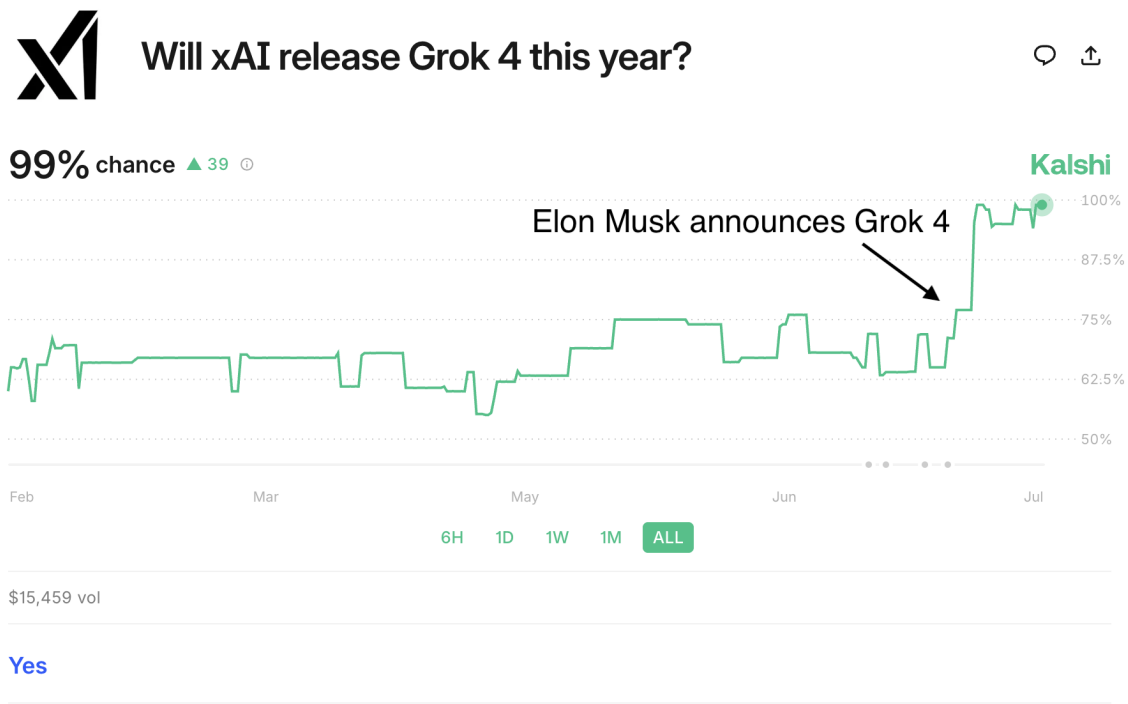


Figure 2: Example of news affecting the market price of a prediction market.

### 5.1 Search Is Critical for Forecasting

In this section, we quantify the impact of search on forecasting performance. As discussed above, timely information is critical for forecasting, and breaking news events can drive large movements in crowd forecasts. Furthermore, this timely information can only be incorporated in LLM-based forecasters via external tools, as the underlying models are retrained infrequently. This induces so-called "knowledge cutoffs," at which point the LLM's knowledge is effectively frozen. For example, OpenAI reports a knowledge cutoff of May 31, 2024 for its flagship o3 reasoning model (OpenAI, 2025), which means the base model has no knowledge of events occurring after this date.[10]

We show that high-quality search dramatically improves forecasting performance, and that prior results to the contrary are partially explained by relatively simple, non-adaptive search pipelines. Second, we show that this effect is strongly mitigated by the practice, which is common in the literature, of directly providing prediction market prices in forecasting prompts. These market prices effectively summarize large volumes of relevant information, and can thus compensate for naive or no-search pipelines.

**Agentic search improves forecasting performance.** We first conduct a series of experiments showing that our agentic search pipeline outperforms a no-search baseline. In order to isolate the effect of search on forecaster performance, we omit the consistency analysis and Platt scaling. We further evaluate two industrial-scale search providers, which

10. The notion of a single "knowledge cutoff" is a simplification (Cheng et al., 2024), but nonetheless is a good heuristic for understanding where the model's general knowledge ends.

we anonymize as Search-A and Search-B. While both providers enable general-purpose semantic search over the internet, Search-A focuses primarily on serving natural language user queries, while Search-B is intended to integrate directly with agentic AI systems. We query both providers via their APIs using both "agentic" and "non-agentic" search pipelines. In the former, the LLM-based forecaster can independently define and execute a query strategy, asking follow-up questions as needed. In the latter, the researcher defines the search pipeline, first prompting the forecaster to generate a handful (here, three) search queries and then providing the results as context to subsequent LLM generations. This latter non-agentic strategy is similar to the one used in Halawi et al. (2024), Karger et al. (2024), and Tao et al. (2025).

As shown in Table 3, our agentic search pipeline with access to the Search-A API and our internal corpus of high-quality news data significantly outperforms a no-news baseline on ForecastBench (0.11435 to 0.1230).

| **Search Method** | **Brier score** |
|---|---|
| None | 0.1230 |
| Search-B (non-agentic) | 0.12168 |
| Search-B (agentic) | 0.11824 |
| Search-A (non-agentic) | 0.11738 |
| Search-A (agentic) | **0.1140** |

Table 3: A comparison of different search strategies. Agentic search outperforms non-agentic search, which in turn only modestly outperforms a no-search baseline.

We also consistently find that, conditional on the source of news, agentic search significantly outperforms non-agentic search, and conditional on the search methodology, Search-A consistently outperforms Search-B. We conjecture that this is due to the relatively low density of relevant information in the typical web page returned by Search-B, while Search-A offers direct responses to questions that are narrowly tailored to the event in question.

We also find that the weakest search method (non-agentic queries to Search-B) performs comparably to simply omitting the retrieval pipeline entirely on ForecastBench. This is consistent with Karger et al. (2024) and Schoenegger et al. (2024), which find no evidence that search improves forecasting performance. However, when viewed in light of our results, this simply highlights the fact that effective search is a nontrivial problem.

**The impact of timely news is mediated by market prices.** We now show that the effects of search are strongly mitigated by providing the forecaster with access to market prices. This fact provides a partial explanation for prior work which suggests that search does not meaningfully impact forecasting performance (Karger et al., 2024; Schoenegger et al., 2024).